SMART BYPASS CHARGER: A USB-PD BASED CHARGING COMPATIBILITY AND MONITORING SYSTEM

Embedded Programming Project Report

Submitted by

S. T. Mithun

Nanthiswaran S. P

Department of Computer Science and Engineering

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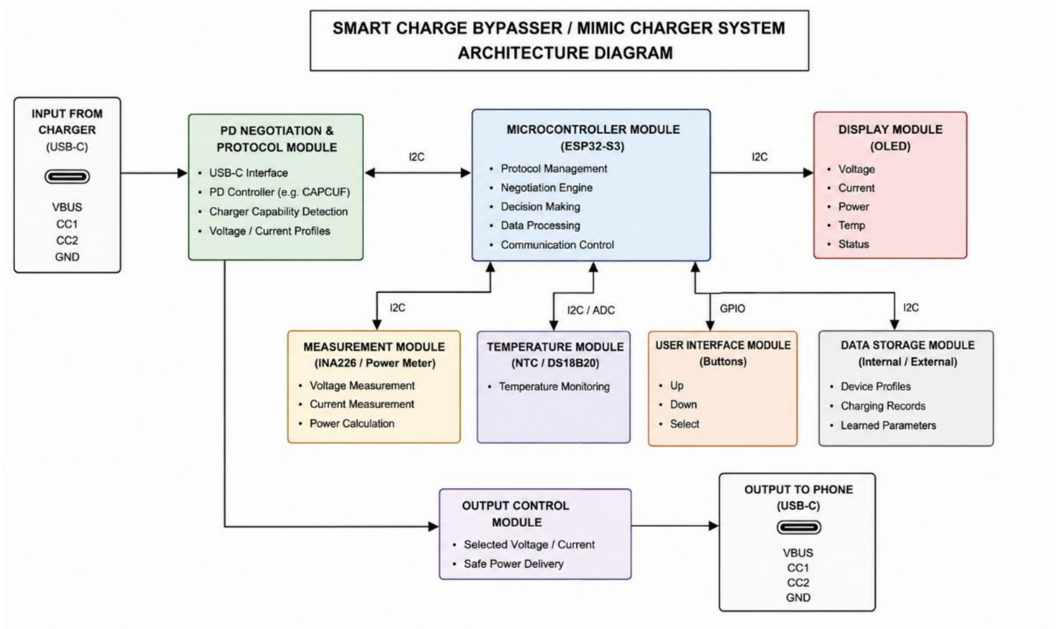
ABSTRACT

Modern smartphones support multiple fast-charging technologies, but charging performance is often reduced when chargers and phones from different brands are used together. This project proposes a Smart Bypass Charger that acts as an intelligent interface between the charger and the smartphone. The system negotiates USB Power Delivery (USB-PD), monitors charging parameters such as voltage, current, power, and temperature, and selects the most suitable charging profile within supported standards. An ESP32 microcontroller controls the charging process, while an INA226 sensor and OLED display provide real-time monitoring and user feedback. The system also maintains charging profiles for future optimization through firmware updates. The proposed solution aims to improve charging compatibility, enhance user awareness, and promote safer and more efficient charging.

KEYWORDS:

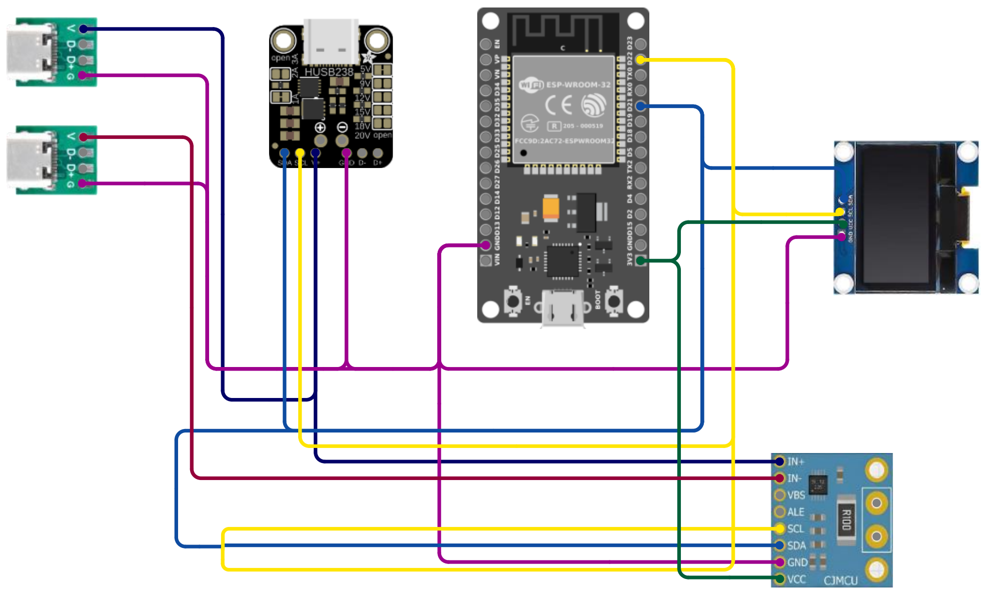
- Smart Charging
- USB Power Delivery (USB-PD)
- ESP32
- USB-C
- Charging Compatibility
- Power Monitoring
- INA226 Sensor
- OLED Display
- Charging Profile Management
- Fast Charging

LOGICAL DESIGN



CIRCUIT DIAGRAM

Cirkit Designer



PHYSICAL LAYOUT